GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 5

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Recap of Previous Lecture

Topic $A \to B$

Topic $B \to A$

Topic $A \equiv B$

Topic

Topic

Turn on Slide map

Topics to be Covered

Topic Converse

Topic Inverse

Topic Contrapositive

Topic Questions

Topic

DISCIPLINE
IS DOING
WHAT NEEDS
TO BE DONE,
EVEN IF
YOU DON'T
WANT TO
DO IT.
PICTUREQUOTES.COM

(Rahul present tabhi होगा Jab Sooraj present ho)

$(A \to B)$

- A is true only when B is true.

(A true tabhi hoga Jab B true hoga)

A	B	$A \to B$	$B \to A$	$A \equiv B$
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	1	1	1

$A \equiv B$

- A is true exactly when B is true

(A is true Jab B true है)

A	B
0	0
0	1
1	0
1	1

Converse, Inverse & Contrapositive in Propositional Logic

- In propositional logic, converse, inverse, and contrapositive are terms related to implication statements (of the form P→Q, meaning "if P, then Q"). These variations help in understanding the relationships between statements and their logical equivalents.

$P \rightarrow Q$

$\Rightarrow$ (if P then Q)

Converse, Inverse & Contrapositive in Propositional Logic

- 1. Converse: The converse of an implication $P \rightarrow Q$ is the statement $Q \rightarrow P$.
- It switches the hypothesis and conclusion.

Converse ⇒ $P \rightarrow Q \xrightarrow{\text{Converse}} Q \rightarrow P$

- Original: $P \rightarrow Q$ "If P, then Q."
- Converse: $Q \rightarrow P$ "If Q, then P."
- Note that the converse is not necessarily logically equivalent to the original implication.
- Example: Original: "If it rains, then the ground is wet." $P \rightarrow Q$
- "Converse: "If the ground is wet, then it rains." (The ground could be wet for other reasons, so this is not logically equivalent.)

$\rightarrow Q \rightarrow P$

Converse, Inverse & Contrapositive in Propositional Logic

- Inverse: The inverse of an implication $P \rightarrow Q$ is the statement $\neg P \rightarrow \neg Q$.
- It negates both the hypothesis and conclusion.
- Original: $P \rightarrow Q$ "If P, then Q."
- Inverse: $\neg P \rightarrow \neg Q$ "If not P, then not Q."
- The inverse is also not necessarily logically equivalent to the original statement.
- Example:
- Original: "If it rains, then the ground is wet."
- Inverse: "If it does not rain, then the ground is not wet." (The ground could be wet for reasons other than rain, so this is also not logically equivalent.)

$$(P \to Q \neq \neg P \to \neg Q \equiv Q \to P)$$

P	Q	$P \to Q$	$\neg P \to \neg Q$
0	0	1	1
0	1	1	0
1	0	0	1
1	1	1	1

Converse, Inverse & Contrapositive in Propositional Logic

- 3. Contrapositive: $(P \rightarrow Q \Rightarrow \neg Q \rightarrow \neg P)$
- The contrapositive of an implication $P \rightarrow Q$ is the statement $\neg Q \rightarrow \neg P$.
- It negates and reverses the hypothesis and conclusion.
- Original: $P \rightarrow Q$"If P, then Q."
- Contrapositive: $\neg Q \rightarrow \neg P$"If not Q, then not P."
- The contrapositive is logically equivalent to the original implication. If an implication is true, its contrapositive will also be true, and vice versa.
- Example:
- Original: "If it rains, then the ground is wet."
- Contrapositive: "If the ground is not wet, then it did not rain." (This is logically equivalent to the original statement.)

logically equivalent.

P	Q	$P \to Q$	$\neg Q \to \neg P$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

Question: If the statement "If it rains, then the ground is wet" is true, which of the following statements is equivalent to "It rains implies the ground is wet" (A → B)?

A B

- A) If the ground is wet, then it rains. $B \to A$
- B) If the ground is not wet, then it does not rain. $\neg B \to \neg A$ (Contrapositive)
- C) The ground is wet implies it rains. $B \to A$
- D) If it does not rain, then the ground is not wet. $\neg A \to \neg B$ (inverse)

Question: Which of the following statements correctly represents the contrapositive of the statement "A implies B"?

$\neg B \rightarrow \neg A$

- A) A implies not B
- B) B implies A
- C) Not B implies not A
- D) Not A implies not B

Question: If the statement "If a number is even, then it is divisible by 2" is given, which of the following is **NOT** an implication of this statement?

A $\quad A \to B \quad$ B

- A) If a number is not even, then it is not divisible by 2. $\neg A \to \neg B$
- B) If a number is divisible by 2, then it is even. $B \to A$
- C) If a number is not divisible by 2, then it is not even. Yes $\neg B \to \neg A$
- D) A number is even if and only if it is divisible by 2. $A \equiv B$

Question: Which of the following is logically equivalent to "A if and only if B" (A ↔ B)?

- A) A implies B and B implies A
- B) A implies B or B implies A
- C) Not A implies not B
- D) A and B are true

A	B	A and B are true	A ↔ B
0	0	0	1
0	1	0	0
1	0	0	0
1	1	1	1

$A \to B$

Question: If the statement "If I study, then I will pass the exam" is true, which of the following is the converse of this statement?

- A) If I do not study, then I will not pass the exam.
- $B \to A$
- B) If I pass the exam, then I studied.
- C) If I do not pass the exam, then I did not study.
- D) I will pass the exam if and only if I study.

imp**

Statement	Form
Original Implication	$P \rightarrow Q$
Converse	$Q \rightarrow P$
Inverse	$\neg P \rightarrow \neg Q$
Contrapositive	$\neg Q \rightarrow \neg P$

Artificial Intelligence

Choose the correct choice(S) regarding the following proportional logic assertion S:

S: $((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$

done

(A) S is neither a tautology nor a contradiction.

(B) S is a tautology.

(C) S is a contradiction.

(D) The antecedent of S is logically equivalent to the consequent of S.

Artificial Intelligence

Let p, q, r denote the statements "It is raining", "It is cold", and "It is pleasant", respectively. Then the statement "It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold" is represented by done

(A) $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$

(B) $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$

(C) $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$

(D) $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

Artificial Intelligence

Let p, q and r be propositions and the expression $(p \to q) \to r$ be a contradiction. Then, the expression $(r \to p) \to q$ is

(A) a tautology

(B) a contradiction

(C) always TRUE when p is FALSE

(D) always TRUE when q is TRUE

	P	Q	R
F	0	0	0
T	0	1	0
T	1	1	0

False

$(P \to q) \to R$

1 0

0 0

0 1

1 1

$(R \to P) \to Q$

1) $(0 \to 0) \to 0 \Rightarrow 0$

$1 \to 0$

2) $(0 \to 0) \to 1 \Rightarrow 1 \to 1 \Rightarrow 1$

3) $(0 \to 1) \to 1 \Rightarrow 1 \to 1 \Rightarrow 1$

Artificial Intelligence

The binary operator $\neq$ is defined by the following truth table.

p	q	p≠q
0	0	0
0	1	1
1	0	1
1	1	0

Which of the following is true about the binary operator $\neq$?

(A) Both commutative and associative
(B) Commutative but not associative
(C) Not commutative but associative
(D) Neither commutative nor associative

q	p	q≠p
0	0	0
0	1	1
1	0	1
1	1	0

Commutative

$(P \neq Q) \neq R \equiv (Q \neq R) \neq P$

P	Q	R	P≠Q	(P≠Q)≠R	(Q≠R)	P≠(Q≠R)
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	1	1	1	1
0	1	1	1	0	0	0
1	0	0	1	0	0	0
1	0	1	1	1	1	1
1	1	0	0	0	1	0
1	1	1	0	1	0	1

Artificial Intelligence

Which one of the following Boolean expressions is NOT a tautology?

done

(A) $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$

(B) $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$

(C) $(a \wedge b \wedge c) \rightarrow (c \vee a)$

(D) $a \rightarrow (b \rightarrow a)$

Artificial Intelligence

The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

I. $p \Rightarrow q$ → False $P=1, Q=0$

II. $q \Rightarrow p$ → False $Q=1, P=0$

III. $(\neg q) \vee p$ → False $\neg Q=0, Q=1, P=0$

IV. $(\neg p) \vee q$ → False $\neg P=0, P=1, Q=0$

(A) I only

(B) I and IV only

(C) II only

(D) II and III only

$\neg P \rightarrow \neg Q$

• False

$\neg P \Rightarrow 1 \Rightarrow P=0$

$\neg Q \Rightarrow 0 \Rightarrow Q=1$

P	Q
0	1

Consider the following logical inferences.

I_1: If it rains then the cricket match will not be played.

A B

The cricket match was played. B=0

Inference: There was no rain. A=0

done

I_2: If it rains then the cricket match will not be played.

- It did not rain. A=0

B=0

- Inference: The cricket match was played.

Which of the following is TRUE?

(A) Both I_1 and I_2 are correct inferences

(B) I_1 is correct but I_2 is not a correct inference

(C) I_1 is not correct but I_2 is a correct inference

A→B is given

1, 2, 4th Row

Case 1

- B=0 is given then A→

	A	B	A→B
1	0	0	1
2	0	1	1
~~3~~	~~1~~	~~0~~	~~0~~
4	1	1	1

Case 2

A=0, Here B Can be 1 or 0.

Consider the following logical inferences.

I_1: If it rains then the cricket match will not be played.

A $\quad A \to B \quad$ B

The cricket match was played.

Inference: There was no rain.

I_2: If it rains then the cricket match will not be played.

It did not rain.

Inference: The cricket match was played.

Which of the following is TRUE?

(A) Both I_1 and I_2 are correct inferences

(B) I_1 is correct but I_2 is not a correct inference

(C) I_1 is not correct but I_2 is a correct inference

$A \to B$ is given

$\neg B \to \neg A$ Contrapositive

Same

~~$\neg A \to \neg B$ Inverse~~

(D) S_1 is not equivalence to S_2

Q10 $\overbrace{(p \to (q \to r))}^{A} \leftrightarrow \overbrace{((p \to q) \to r)}^{B}$ is

(A) Tautology

(B) Contingency

(C) Contradiction

(D) Can't be determined.

1 0

$P \to (Q \to R)$

1 0

False 110

$(P \to q) \to R$

1 0

000

010

110

P Q R	A	B	A↔B
000	1	0	0
001	1	1	1
010	1	0	0
011	1	1	1
100	1	1	
101	1	1	
110	0	1	
111	1	1	

$S_1 : \sim(p \vee (\sim p \wedge q))$

$S_2 : \sim p \wedge \sim q$

Which of the following is correct?

(A) S_1 is tautology

(B) S_2 is contradiction

(C) S_1 is equivalence to S_2

(d) S_1, S_2 not equivalent

S_2	P	Q	S_1
1	0	0	1
0	0	1	0
0	1	0	0
0	1	1	Q

$S_1 = \sim(P \vee (\sim P \wedge Q))$

If $P=1$ $S_1 = \sim(1) = 0$

If $P=0$ $S_1 = \sim(1 \wedge Q) = \sim Q$

$S_2 = \underbrace{\sim P} \wedge \underbrace{\sim Q}$

$S_2 = 1$ when $\sim P = 1$, $\sim Q = 1$ $\Rightarrow P = Q = 0$

Q7 Which of the following is/are logical equivalence?

I. ~ (p → q)

II. ~ (p → q) ∧ (q → r)

III. p ∧ ~q

IV. (p ∨ q) → r

(A) I and II (B) I and III

(C) II and IV (D) II and III

Q8 Consider the following statement

S_1 : ~ (p ↔ q)

S_2 : p ↔ ~q

Which of the following is correct?

(A) S_1 is tautology

(B) S_2 is contradiction

(C) S_1 is equivalence to S_2

(D) None of these

$P \wedge \sim Q$ false

P	Q	R
0	0	0
0	1	0
1	1	0
0	0	1
0	1	1
1	1	0

I, II, III

i) $\sim(P \to Q)$ false

P	Q	R
0	0	0
0	1	0
1	1	0
0	0	1
0	1	0
1	1	0

True (100, 101)

ii) $\sim(P \to Q) \wedge (Q \to R)$ false

0 0 1 0

0 1

1 1

P	Q	R
0	0	0
0	1	0 ✓
1	1	0 ✓
0	0	1
0	1	1
1	1	1

$(P \vee Q) \rightarrow R$ false

1 0

P	Q	R
0	1	0
1	0	0
1	1	0

THANK - YOU